

Micro-teaching Experience #1:

Micro-teaching Experience # 1 is to engage peers or advanced professionals in finding better solutions for their technical problems. It will be offered for different companies in the logistics/supply chain field.

1

They are a senior engineer in the supply chain division. They use classical modeling tools but feel that they have little time to update their toolbox with new solutions. They are motivated to gain new skills and network with other peers in different companies.

2

They are a senior project manager of the logistics department. They have good knowledge of OR tools of the use of OR tools but hardly implement them. They want to understand how they can improve their communication with the engineers working in their department regarding tool decisions.

It will be an online series of workshops, where the learners will be presented with an overview of newer algorithmic strategies and a discussion of the limitations and benefits of each one. The workshops will focus on the technical aspects of the algorithms and practical application. Each will present a new technique that starts with the big-picture overview and sharing personal experiences with the problems and tools I am teaching, and it will end with a discussion of the limitations and benefits of the tool. Also, we will discuss which other solutions or changes can improve the solution. Engagement will be incentivized at the beginning of the class by presentations of personal experiences in the field. Each session will last 3 hours, there will be 4 sessions in the span of six months to one year.

Their motivation will be to learn new skills and learn to improve specific advanced technical modeling skills and network with peers. They can choose the events they will participate in based on their skills gaps.

The challenge is to assess and address common misconceptions of the algorithmic solutions to OR problems. Also, the level of technical skill will vary, which is why I am proposing a series of different tools, so they can pick and choose the ones that are more relevant.

Micro-teaching Experience #2:

Micro-teaching Experience # 2 is to engage novice professionals in your field in critical thinking regarding the choices they make in modeling and research and their consequences in the real world.

1

They are junior researchers in a start-up company. They are interested in pure optimization and data science, especially AI. They have good knowledge of OR tools taught in school, but they still have trouble seeing the big picture and transforming the company's business challenges into solvable models. They also don't know many people in their field and think that this workshop will help them increase their network.

2

They are PhD students in the Industrial Engineering field. They are still working in their first paper, but they want to get more tools on modelling and social responsibility. This workshop will help them to hone their

critical thinking that will be essential for a successful career in Academia.

It will be a 1.5-hour in-person workshop, followed by 1 hour lunch. They will also be connected online in a group chat, where they will be encouraged to use throughout the day, to engage them in connections.

They are going to be presented with different modelling scenarios that are prone to instigate ethical inquiries. Engagement will be incentivized at the beginning of the class by presentations of real-world decision-making gone wrong followed by group discussion of what happened. Their motivation will be to improve their knowledge of modeling and decision-making. There is the added bonus to build a community of peers.

The challenge is to focus the class on critical thinking instead of practical mathematical application, common in Engineering courses. I proposed this set-up to encourage discussion and peer-to-peer inquiries.

Micro-teaching Experience #3:

Experience meant engaging the public not in the field.

1

The first person is a stay-at-home parent that uses the library to bring their kids and build community. They use the library resources regularly for self-improvement. They are curious about the possibility of having a tool that can improve their day-to-day life.

2

They are a young professional. He is aware of the process of solving problems called Design thinking, which have similarities with modeling. He is curious about using data to improve decision making.

The artifact is going to be an online game with at least 5 levels, which might be completed in an hour or two. There will be no knowledge of math required. The idea is to reach an understanding of how to use OR strategies to improve decision-making when there are limited resources. To motivate learners, I am using routine/household problems as each of the case studies - e.g.: scheduling activities in a busy calendar. There will be a brief explanation before each level about the mechanism behind the game (OR tools). and we will prompt discussions on the concepts of feasibility x optimality concepts and how to transfer this knowledge into their lives after they passed each level.

The game will be an asynchronous web resource provided by the public library to engage the public. The learners will range in age, since even kids will be allowed to play, with diverse backgrounds. Their main goal is skill build and having fun. Only the library patrons will have access to resources, and link to contact instructors will be offered.

The gamification will offer the motivation to continue playing and interacting with the material. There will not be a habit, but the person would be able to play again without the same exact game, since there will be randomness associated with the game engine.

The main challenge is to explain simply but effectively a complicated concept that sometimes baffles advanced undergraduate in engineering courses (feasibility x optimality). The idea of the game is to be able to be helpful and clear to that diverse audience, without being ineffective, or too easy, which is tricky.

Besides, it is hard to make people interested in accessing online resources with so many competing apps/games/educational materials. Also, there will be necessary to get resources to design, develop and run the application.

Micro-teaching Experience #4:

Experience meant to focus on another public

1

They are a sophomore in high school student trying to decide which career to pursue (major in college). They are interested in experimenting with different topics and fields before applying to college. They have multiple interests at school and on hobbies. They are complete beginners though when it comes to Operations Research but are interested in learning more.

2

They are a junior in high school interested in mathematical applications. They have heard about Operations Research, but they never took a course on it. They might have some misconceptions regarding the real-world application and capabilities of the tools.

Outreach experiences for high schoolers with 2 hour long experiential learning experience with a 15-minute break in the middle. The students have probably no knowledge of OR is, and very limited knowledge of the applied mathematics field.

For this experience, I am planning a experiential experience in the format of game/workshop. There will be a brief presentation on what OR professionals do and presentation of the case study. We then will divide the students into groups of 5, each will work in the same case but each in a station, where there will be an independent but connected simple task, part of the whole real-world problem they are instigated to solve. Besides the practical aspect of the exercise, each small simple activity is going to be planned to be 10 minutes long. It will have a game mechanism, with randomness associated with each station. At the end of each session there will be a 5-minute debriefing. At the end of the activity, there will be a class discussion of each station, and I am revisiting the case study to compare notes on the problem solutions and results.

The objective is to teach the mechanics of decision-making using mathematical models but not the mathematical or technical aspects of it. We want them to be able to name the main challenges of the activities, and the effect of decisions and randomness in the outcome.

The main challenge is to make it fun and simple but not detached from the case study problem. Each station has to engage the students and teach a different part of the problem; this means that the problem will need to be complex enough to offer diverse material. Besides, since I am dividing the class into 5 stations means we need at least 5 volunteers' specialists to help with the experience.